ALICE L WALTON SCHOOL OF MEDICINE

Hemodynamics and Shock: Foundations and Frameworks

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Authors: Chelsey Deel, Stephen Nix
Editor: Jordan Finn

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Learning Objectives (4)

After completing this brick, you will be able to:

1. Describe the clinical manifestations and pathophysiologic categories of edema.
2. Compare exudate vs. transudate.
3. Compare and contrast hyperemia and congestion.
4. Differentiate

CASE CONNECTION

A 50-year-old male, JG, presents to your clinic for a follow-up visit. Review of his chart shows that he has liver cirrhosis secondary to alcohol use disorder. During his physical exam, you notice he has a large, protuberant abdomen. JG remarks, "The doctors have told me I have a lot of fluid in my belly. Where did that fluid come from?"

How would you answer JG's question? Consider this as you read, and we will revisit his case at the end of the brick.

GO TO CONCLUSION ↓

Edema and Effusions (LO #1)

Under normal conditions, the body's fluid distribution is governed by a balance of opposing forces (so-called Starling Forces): **hydrostatic pressure** which tends to push fluid out of the blood vessel and **plasma colloid osmotic pressure** which tends to pull fluid into the vessel (see Figure 1). There is usually a small net movement of fluid into the interstitial space; however, this typically drains into the lymphatic vessels and ultimately returns to the bloodstream.

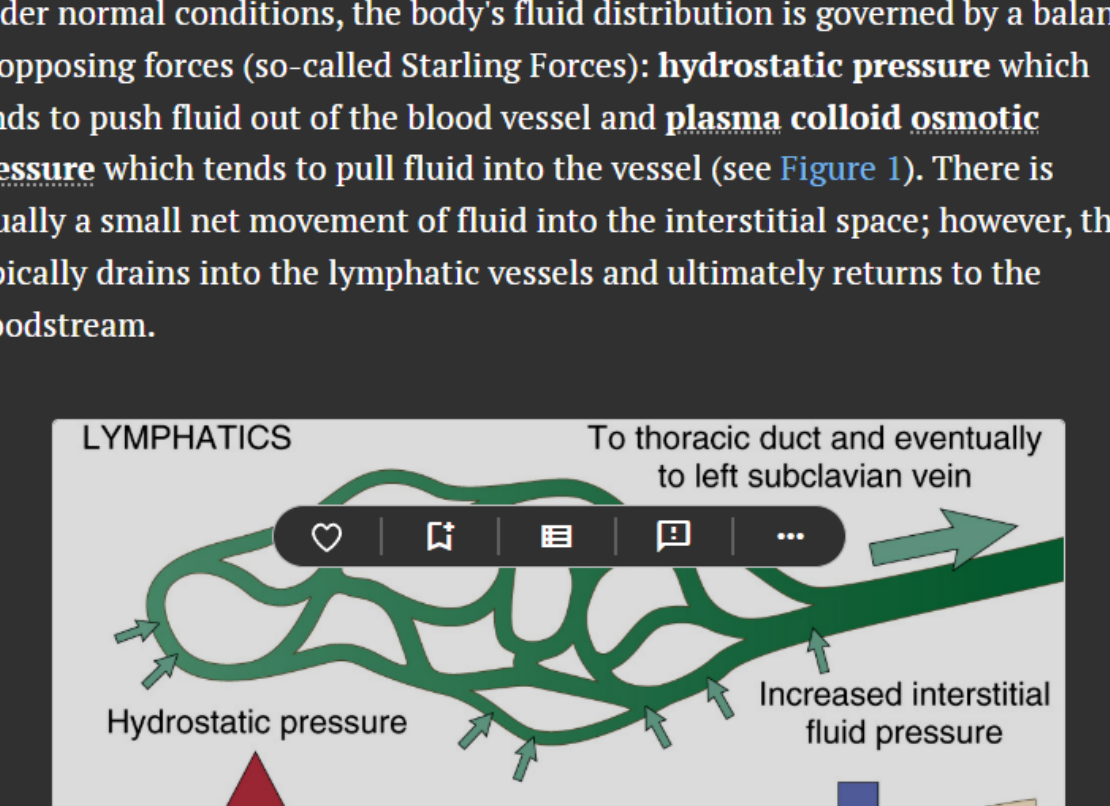
LYMPHATICS
To thoracic duct and eventually to left subclavian vein
Hydrostatic pressure
Increased interstitial fluid pressure
Plasma colloid osmotic pressure
Arterial end
CAPILLARY BED
Venous end

Figure 1

When those forces are disturbed, we start to accumulate fluid in tissues or body cavities. Here are some important terms to illustrate clinical manifestations of edema:

- **Edema:** accumulation of fluid in tissues
- **Effusion:** accumulation of fluid within a body cavity
  - Pleural cavity- **pleural effusion**
  - Pericardial sac- **pericardial effusion**
  - Abdominal cavity- **ascites**
- **Dependent (pedal) edema:** edema of the lower extremities
- **Anasarca:** generalized edema
- **Lymphedema:** e

Pathophysiologic Causes of Edema (LO #1)

How edema develops is typically a function of altering those forces that govern fluid balance.

Increased Hydrostatic Pressure

Increases in hydrostatic pressure are predominantly due to disorders that impair venous return. Examples of this are congestive heart failure, constrictive pericarditis, and liver cirrhosis, as it obstructs the portal vein.

CLINICAL CORRELATION

If the impairment is localized (such as a deep venous thrombosis, or DVT, in the leg), the resultant edema is usually confined to that body part. The disorders mentioned above cause a more systemic increase in venous pressure and lead to more widespread edema.

Reduced Plasma Colloid Osmotic Pressure

The predominant plasma protein that contributes to osmotic pressure is albumin (almost 50% of total plasma protein); therefore, conditions in which there is a loss of albumin cause a reduced osmotic pressure. Examples include severe liver disease/cirrhosis, protein-losing glomerulopathies in the kidney (nephrotic syndrome teropathies.

Sodium and Water Retention

Sodium retention leads to water retention, as "water follows sodium". This leads to an increased hydrostatic pressure due to intravascular fluid volume expansion and diminished osmotic pressure due to dilution. This mainly occurs when kidney function is compromised, either due to a primary kidney disorder or a cardiovascular disorder such as heart failure that causes renal hypoperfusion.

Lymphatic Obstruction

Movement of fluid from the interstitial space into lymphatic vessels can be blocked by various conditions, including trauma, fibrosis, invasive tumors, and infectious diseases such as filariasis from parasitic infection by the Filarioidea roundworm (Figure 2).

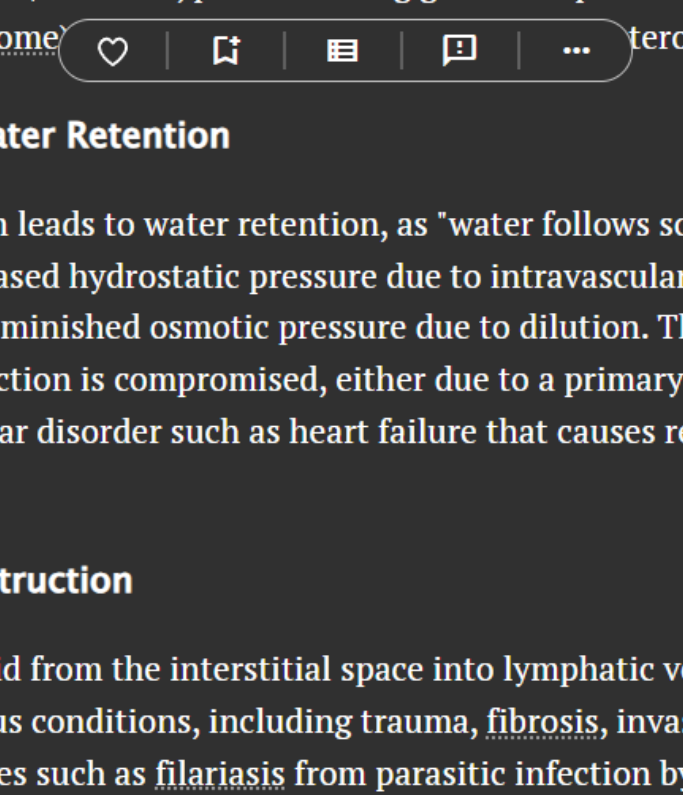

Figure 2: Massive edema and elephantiasis caused by filariasis of the leg

Increased Vascular Permeability

During times of inflammation, the effect of inflammatory cytokines on vessel walls causes them to have increased permeability and allow more efflux of fluid into the interstitium.

Additionally, new vessels formed in a process known as angiogenesis are leakier than mature vessels, due to incomplete interendothelial junctions and vascular endothelial growth factor (VEGF) causing more vascular permeability.

Name the pathophysiologic categories of edema.

CLINICAL CORRELATION

Clinical consequences of edema depend on the extent and location. Subcutaneous edema, though not immediately life-threatening, may be a sign of underlying cardiac or renal dysfunction. Pulmonary edema impairs gas exchange, leading to hypoxemia, and provides a fertile ground for infection (pneumonia). Edema in the brain is life threatening. Because the skull is a limited physical space, severe brain edema can cause herniation of brain tissue through spaces like the foramen magnum, which can press on centers in the brainstem that control cardiac and respiratory function.

Exudates vs. Transudates (LO #3)

Edema and effusions can be inflammatory and non-inflammatory (Figure 3).

Exudates are inflammatory fluid collections that arise due to increases in vascular permeability caused by inflammatory mediators. These collections are protein-rich with a higher cellularity and specific gravity. Common disorders include infections (pneumonia, tuberculosis), inflammatory disorders (lupus, rheumatoid arthritis), and malignancy.

Transudates are non-inflammatory fluid collections that are common in many disorders, such as heart failure, liver failure, renal disease, and malnutrition. They are essentially ultrafiltrates of plasma, with low levels of protein, cellularity, and specif

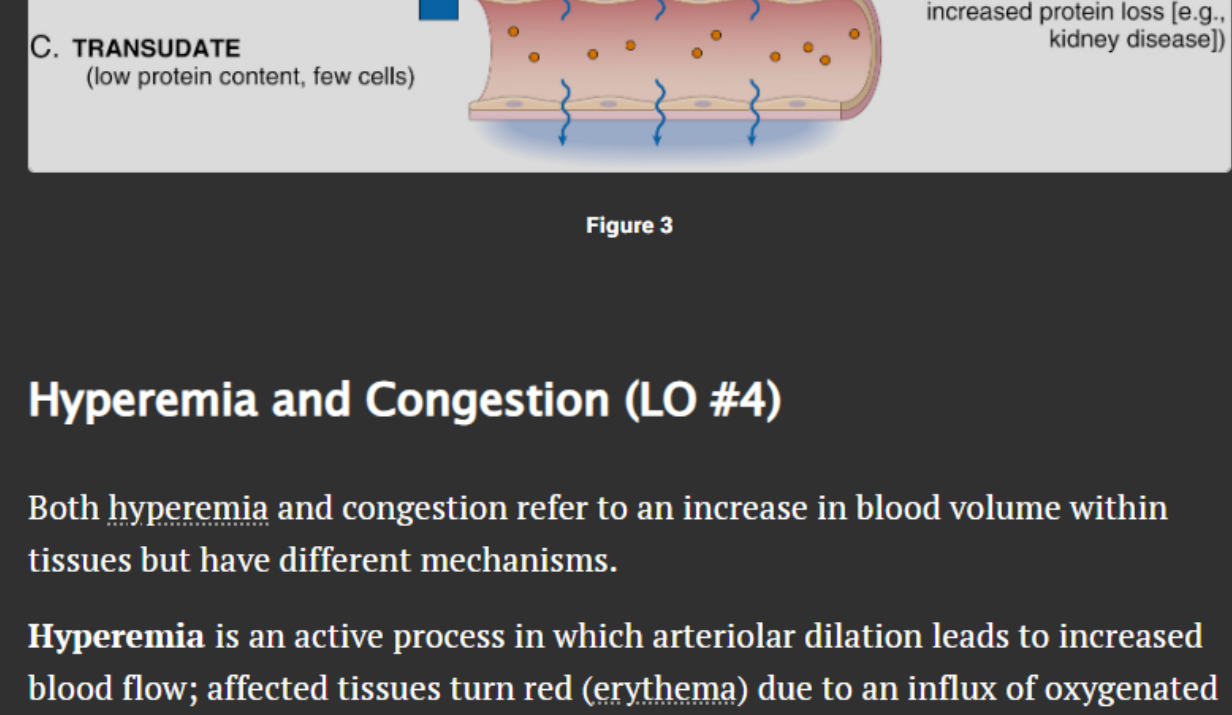
A. NORMAL
Hydrostatic pressure
Colloid osmotic pressure
Plasma proteins
B. EXUDATE (high protein content, and may contain some white and red cells)
Fluid and protein leakage
Vasodilation and stasis
Increased interendothelial spaces
Inflammation
Increased hydrostatic pressure (venous outflow obstruction, [e.g., congestive heart failure])
C. TRANSUDATE (low protein content, few cells)
Fluid leakage
Decreased colloid osmotic pressure (decreased protein synthesis [e.g.,liver disease]; increased protein loss [e.g., kidney disease])

Figure 3

Hyperemia and Congestion (LO #4)

Both hyperemia and congestion refer to an increase in blood volume within tissues but have different mechanisms.

Hyperemia is an active process in which arteriolar dilation leads to increased blood flow; affected tissues turn red (erythema) due to an influx of oxygenated blood. This can arise e, or other stimuli.

Congestion is a passive process resulting from reduced venous outflow of blood from a tissue. It can be systemic, as in cardiac failure, or localized, as in isolated venous obstruction. Congested tissues may take on a cyanotic (reddish-blue) appearance due to accumulation of deoxygenated hemoglobin. Over time, chronic congestion can lead to ischemic tissue injury and fibrosis.

Congestion or Hyperemia?: A 48-year-old male develops a blood clot in his right popliteal vein. His right lower leg becomes swollen and discolored.

Congestion or Hyperemia?: A 28 year old man is unexpectedly called on during a meeting and starts to blush.

Shock (LO #5)

Shock is a term that you may have heard of in popular culture but may not know the exact mechanism. The various causes will come up again in the systems courses, but we will provide a general overview in this brick.

Shock is a state of circulatory failure in which there is an inability to supply blood and oxygen to organs and tissues. The cellular injury may become irreversible and even fatal if the process is prolonged. The following are the the 5 main types of shock:

- Hypovolemic shock: results from low cardiac output due to low blood volume (ex. massive hemorrhage or fluid loss from severe burns)
- Cardiogenic shock: results from low cardiac output due to heart pump failure (ex. myocardial infarction/heart attack, arrythmias, etc.)
- Septic shock: results from states of systemic inflammation (overwhelming bacterial or fungal infections, toxic shock syndrome, burns, pancreatitis, etc.)
- Neurogenic shock: results from spinal cord injury (acute vasodilation leads to hypotension and tissue hypoperfusion)
- Anaphylactic shock: IgE mediated hypersensitivity reaction to an allergen (acute vasodilation leads to hypotension and tissue hypoperfusion)

CLINICAL CORRELATION

The clinical manifestations of shock depend on the precipitating insult. For hypovolemic and cardiogenic shock, patients exhibit hypotension, a weak rapid pulse, tachypnea, and cool/clammy/cyanotic skin. Patients with septic shock may have warm and flushed skin due to peripheral vasodilation.

CASE CONNECTION

BACK TO INTRODUCTION ↑

After finishing your physical exam with JG, you have a lengthy discussion about his liver disease. "Unfortunately, because your liver is not functioning as it should, a few things happen. Your liver is responsible for making important proteins that keep fluid in your blood; because you have less protein, fluid remains in the tissues and body cavities. Additionally, cirrhosis is like having scar tissue in your liver. Blood does not flow as easily through that scar and backs up in your bloodstream, pushing more fluid out of your blood vessels."

JG thanks you for the explanation and looks forward to hearing about possible interventions for his chronic liver disease.

Summary

- Edema and effusions occur when fluid distribution in the body is disturbed, leading to accum ties.
- Pathophysiologic causes of edema include increased hydrostatic pressure, reduced plasma colloid osmotic pressure, sodium and water retention, lymphatic obstruction, and increased vascular permeability.
- Edema and effusions can be inflammatory (exudates) or non-inflammatory (transudates). Exudates are protein-rich with higher cellularity, while transudates are ultrafiltrates of plasma with low levels of protein and cellularity. Examples of exudates include infections (pneumonia, tuberculosis), inflammatory disorders (lupus, rheumatoid arthritis), and malignancy. Examples of transudates include heart failure, liver failure, renal disease, and malnutrition.
- Clinical consequences of edema depend on its extent and location. For instance, subcutaneous edema may indicate cardiac or renal dysfunction, pulmonary edema can lead to hypoxemia and infection, and brain edema can be life-threatening.
- Hyperemia and congestion both refer to an increase in blood volume within tissues. Hyperemia is an active process due to arteriolar dilation, while congestion is a passive process resulting from reduced venous outflow. Examples of hyperemia include blushing and increased blood flow to areas of exercise and injury/infection. Examples of congestion include chronic pulmonary and hepatic congestion secondary to congestive heart failure.
- Shock is a state of circulatory failure that can lead to irreversible cellular injury and fatality. Types of shock include hypovolemic, cardiogenic, septic, neurogenic, and anaphylactic shock.

Review Questions

1. Which of the follo f fluid within the peritoneal cavity?

A. ○ Edema
B. ○ Pleural Effusion
C. ○ Anasarca
D. ● Ascites
E. ○ Pericardial Effusion

Hide Explanation

The correct answer is ascites (D). Edema (A) is the general term for accumulation of fluid within tissues. Pleural effusion (B) is fluid accumulation in the pleural cavity. Anasarca (C) is generalized body edema. Pericardial effusion (E) is fluid accumulation in the pericardial sac.

2. A 65-year-old male presents to your clinic for follow-up. You notice the patient has swelling around his eyes (periorbital edema). He mentions to you he has a history of nephrotic syndrome, a kidney disease in which the patient loses a significant amount of protein in the urine. Which of the following categories is contributing to the patient's edema?

A. ● Reduced plasma colloid osmotic pressure
B. ○ Increased hydrostatic pressure
C. ○ Lymphatic obstruction
D. ○ Increased vascular permeability
E. ○ Sodium and water retention

Hide Explanation

The correct answer is reduced plasma colloid osmotic pressure (A). Increased hydrostatic pressure (B) is predominantly due to disorders that impair venous return. Examples of this are congestive heart failure, constrictive pericarditis, and liver cirrhosis, as it obstructs the portal vein. Lymphatic obstruction (C) occurs when movement of fluid from the interstitial space into lymphatic vessels is blocked, such as with trauma, fibrosis, invasive tumors, and infectious diseases. Increased vascular permeability (D) occurs during times of inflammation or new vessel formation (angiogenesis). Sodium and water retention (E) may be occurring in this patient with renal disease; however, the hallmark of nephrotic syndrome is the massive loss of protein in the urine and thus lower levels in the blood, causing a reduced colloid osmotic pressure.

3. Which of the following parameters is compatible with a transudate?

A. ○ Increased cellularity in fluid
B. ○ High protein level
C. ○ History of bacterial infection
D. ● History of heart failure
E. ○ High specific gravity

Hide Explanation

The correct answer is history of heart failure (D). Transudates have low cellularity, not increased cellularity (A). Transudates have low protein levels, not high (B). Exudates, not transudates, occur in a background of infection (C). Transudates have low specific gravity, not high specific gravity (E).

4. Which of the following is true of hyperemia?

A. ○ It is a passive process.
B. ○ Affected tissues take on a cyanotic appearance due to accumulation of deoxygenated hemoglobin.
C. ○ The process can lead to ischemic tissue injury and fibrosis over time.
D. ○ Associated with systemic diseases such as cardiac failure
E. ● Arises as a reaction to inflammation, exercise, or other stimuli

Hide Explanation

The correct answer is that hyperemia arises as a reaction to inflammation, exercise, or other stimuli (E). Hyperemia is an active process, not a passive process (A). Affected tissues take on a red appearance due to an influx of oxygenated blood, not a cyanotic appearance (B). Congestion, not hyperemia, can lead to ischemic tissue injury and fibrosis over time (C). Congestion, not hyperemia, is associated with systemic diseases such as cardiac failure (D).

5. A 25-year-old hiker presents to the ER after being attacked by a mountain lion with large, bleeding wounds on her arms and legs. Her skin is cool and clammy, and she has a heart rate of 140 beats/minute (normal: 80-100 bpm) and blood pressure of 100/60 mmHg (normal: 120/80 mmHg). What type of shock is she experiencing?

A. ○ Anaphylactic
B. ● Hypovolemic
C. ○ Neurogenic
D. ○ Cardiogenic
E. ○ Septic

Hide Explanation

The correct answer is hypovolemic shock (B), secondary to massive hemorrhage. Anaphylactic shock (tion to an allergen (e.g., bee sting). Neurogenic shock (C) occurs in the setting of a spinal cord injury. Cardiogenic shock (D) results from low cardiac output due to heart pump failure (myocardial infarction/heart attack, arrythmias, etc.). Septic shock (E) results from states of systemic inflammation (overwhelming bacterial or fungal infections, toxic shock syndrome, burns, pancreatitis, etc.).

How would you rate this Brick?

References

- Kumar V, Abbas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease.* 10th ed. Elsevier; 2021.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

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03 ☑ ~~Compare and contrast hyperemia and congestion.~~
04 ☑ ~~Differentiate the different types of shock.~~

Objective Confidence: 4/4